Churn Prediction & Retention Strategy (Capstone Project)

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Executive Summary

This project predicts telecom churn using ML. It helps businesses take proactive actions to retain customers.

Problem Statement

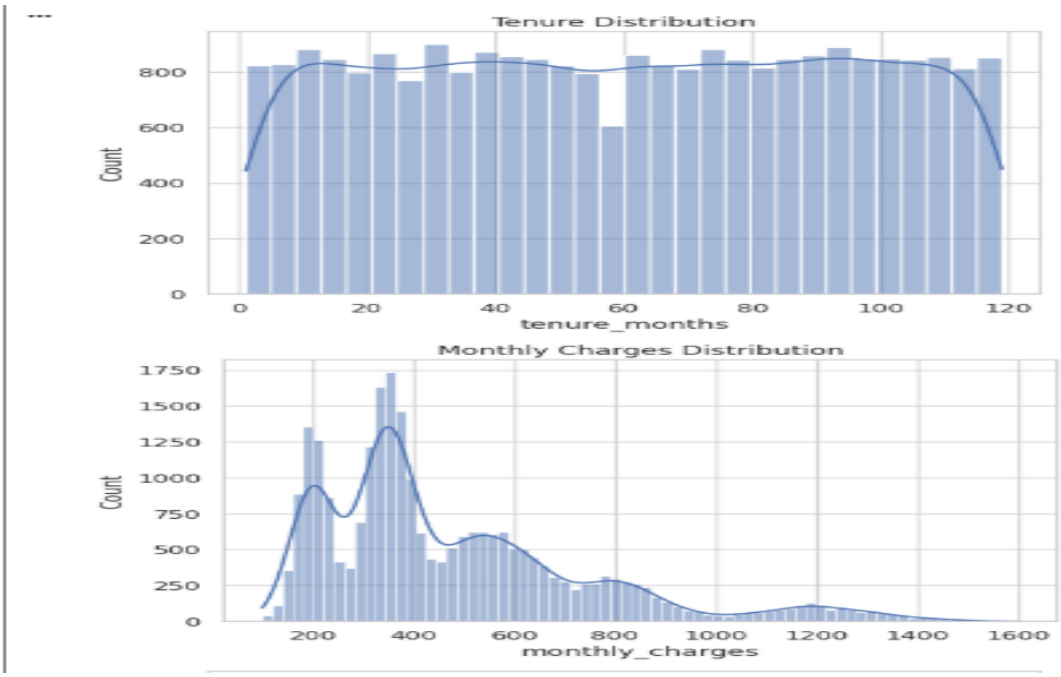
Churn causes revenue loss. The goal is to predict churn and reduce it.

Dataset Overview

Dataset includes demographics, billing, and usage. Target variable is is_churn.

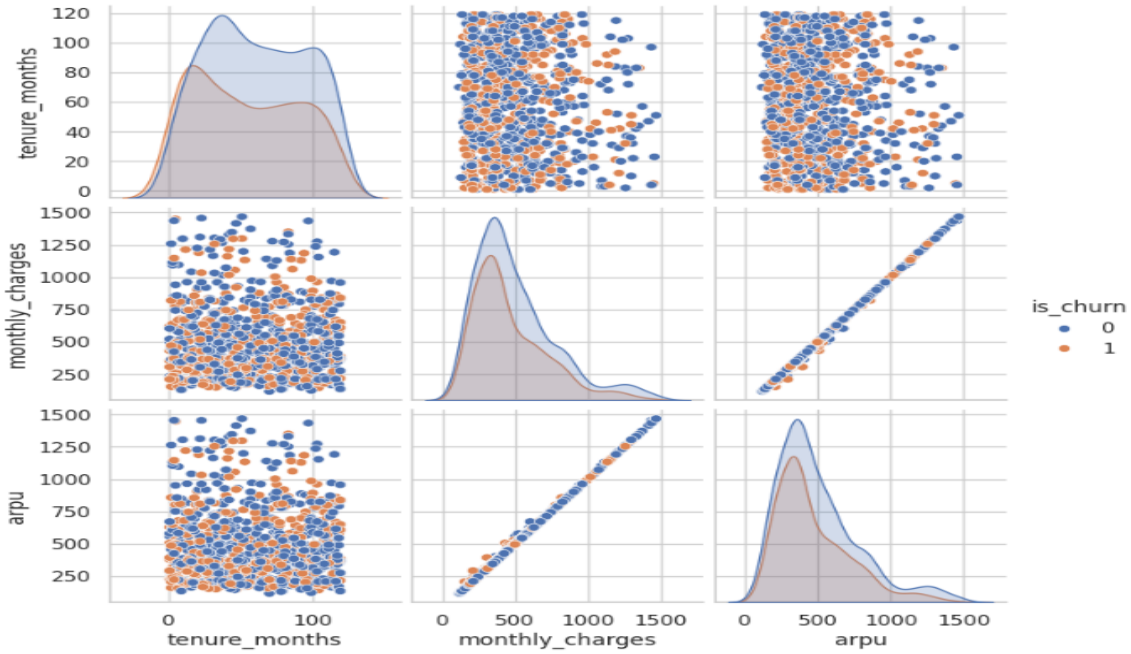
EDA Insights

EDA shows churn is higher for low tenure and high charges.



Advanced EDA

Relationships and correlations reveal key churn drivers.



Feature Engineering

Created new behavioral features to improve predictions.

Model Building

XGBoost pipeline used with preprocessing.

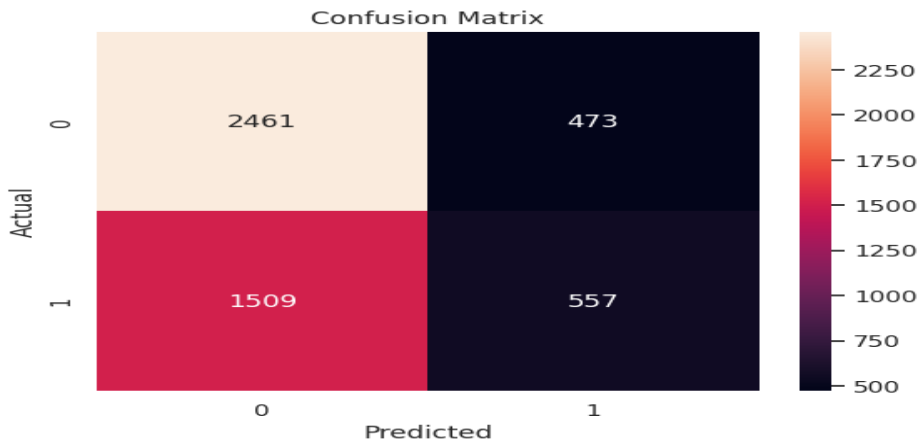
Model Evaluation

Model evaluated using confusion matrix and ROC curve.

Confusion Matrix

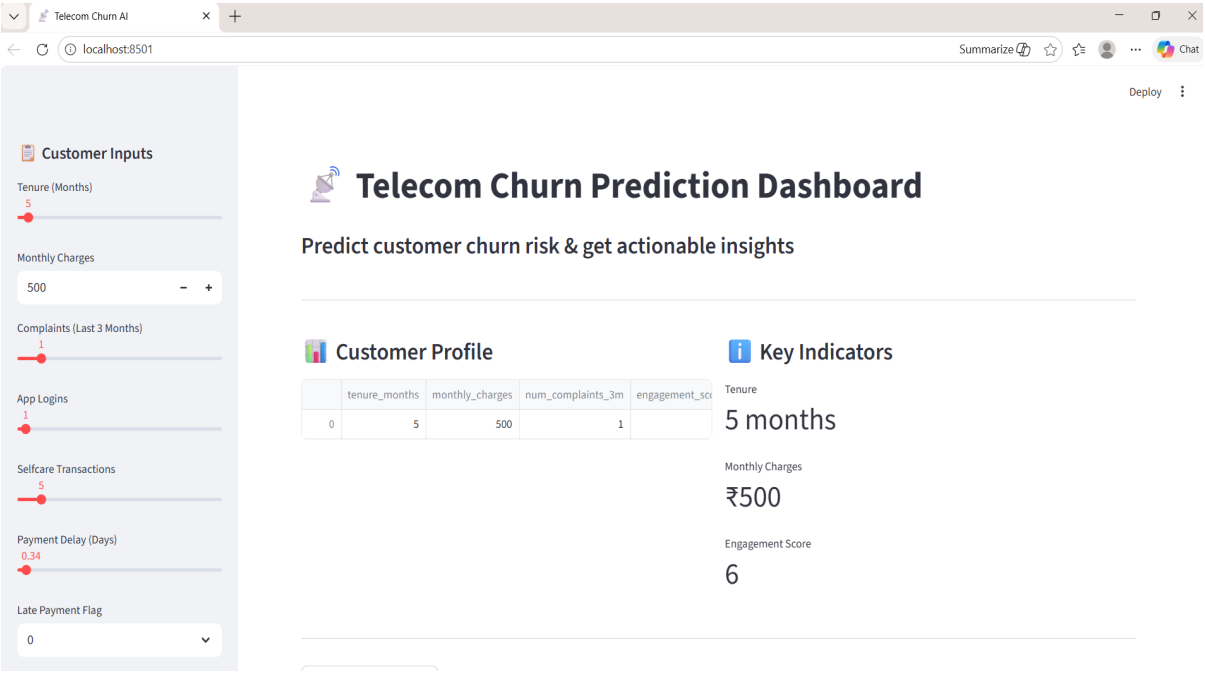
```
cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



Streamlit Application

Interactive app for real-time churn prediction.



Business Recommendations

Target high-risk customers and improve engagement.

Conclusion

ML enables proactive retention and improves revenue.